

# GENERAL RELATIVITY

## Tutorial problem set 4, 02.10.2020, solutions

### ■ PROBLEM 10 Volume element.

The volume element can be written in terms of an  $n$ -form as

$$d^n x \equiv \frac{1}{n!} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_n}, \quad (10.1)$$

where  $\tilde{\epsilon}_{\mu_1 \dots \mu_n}$  is the totally antisymmetric Levi-Civita symbol.

- (a) Show that  $\sqrt{-g}d^n x$  is the invariant volume element under coordinate transformations, where  $g \equiv \det(g_{\mu\nu})$ . Conclude from there that, when  $n = 4$ ,  $\epsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\tilde{\epsilon}_{\mu\nu\rho\sigma}$  is a tensor.
- (b) Can you discuss what happens to the volume element in  $n = 3$  space when you introduce a left-handed instead of right-handed coordinate system? Hint: Consider first a left-handed and right-handed coordinate system on an  $n = 2$  surface embedded in  $n = 3$  space, and a surface element on it, and think of orientability.

**SOLUTION:** (a)  $\sqrt{-g}d^n x$  in (10.1) transforms as,

$$\begin{aligned} \sqrt{-g}d^n x &\rightarrow \sqrt{-g'}d^n x' = \frac{1}{n!} \sqrt{-g'} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx'^{\mu_1} \wedge dx'^{\mu_2} \wedge \dots \wedge dx'^{\mu_n} \\ &= \frac{1}{n!} \sqrt{-g'} \tilde{\epsilon}_{\mu_1 \dots \mu_n} \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \frac{\partial x'^{\mu_2}}{\partial x^{\alpha_2}} \dots \frac{\partial x'^{\mu_n}}{\partial x^{\alpha_n}} dx^{\alpha_1} \wedge dx^{\alpha_2} \wedge \dots \wedge dx^{\alpha_n}. \end{aligned} \quad (10.2)$$

Therefore, we need to show that,

$$\sqrt{-g} \tilde{\epsilon}_{\alpha_1 \dots \alpha_n} = \sqrt{-g'} \tilde{\epsilon}_{\mu_1 \dots \mu_n} \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \frac{\partial x'^{\mu_2}}{\partial x^{\alpha_2}} \dots \frac{\partial x'^{\mu_n}}{\partial x^{\alpha_n}}. \quad (10.3)$$

On the other hand, recall that the determinant can be written as,

$$g = \frac{1}{n!} \tilde{\epsilon}^{\alpha_1 \dots \alpha_n} \tilde{\epsilon}^{\mu_1 \dots \mu_n} g_{\alpha_1 \mu_1} g_{\alpha_2 \mu_2} \dots g_{\alpha_n \mu_n}, \quad (10.4)$$

(check explicitly that in two dimensions this expression indeed returns the determinant) from which we conclude that it transforms as,

$$\begin{aligned} g \rightarrow g' &= \frac{1}{n!} \tilde{\epsilon}^{\alpha_1 \dots \alpha_n} \tilde{\epsilon}^{\mu_1 \dots \mu_n} g'_{\alpha_1 \mu_1} g'_{\alpha_2 \mu_2} \dots g'_{\alpha_n \mu_n} \\ &= \frac{1}{n!} \frac{\partial x^{\nu_1}}{\partial x'^{\mu_1}} \frac{\partial x^{\nu_2}}{\partial x'^{\mu_2}} \dots \frac{\partial x^{\nu_n}}{\partial x'^{\mu_n}} \frac{\partial x^{\beta_1}}{\partial x'^{\alpha_1}} \frac{\partial x^{\beta_2}}{\partial x'^{\alpha_2}} \dots \frac{\partial x^{\beta_n}}{\partial x'^{\alpha_n}} \tilde{\epsilon}^{\alpha_1 \dots \alpha_n} \tilde{\epsilon}^{\mu_1 \dots \mu_n} g_{\beta_1 \nu_1} g_{\beta_2 \nu_2} \dots g_{\beta_n \nu_n} \end{aligned} \quad (10.5)$$

Now, taking a square root of this, we see from (10.3) that the transformation of  $\sqrt{-g}$  is compensated by the transformation of the basis, such that  $\epsilon_{\mu_1 \mu_2 \dots \mu_n} = \sqrt{-g} \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n}$  transforms as a tensor. This tensor is known as the Levi-Civita tensor, and it is used to define the integration measure on manifolds.

- (b) The volume element changes *sign*. That means that the Levi-Civita tensor defines an *oriented volume*, and it can be thought of as a generalization of the cross product defined in three spatial dimensions on flat space.

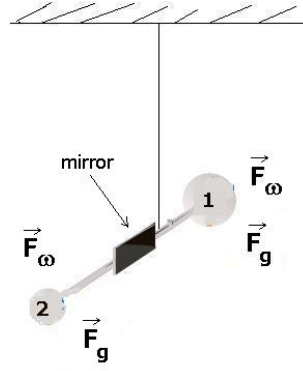


Figure 1: Torsion balance modelling the one used by Loránd Eötvös. Two weights on the ends of the rod are made of different materials. The mirror reflects the light beam such that one can detect the rotation due to the torque.

### ■ PROBLEM 11 The Eötvös experiment.

An experimental set-up in Figure 1 shows an idealized picture of how Eötvös tested the Equivalence principle (*i.e.*  $M_{\text{grav}} = M_{\text{inert}}$ ). The rod in the laboratory frame (the one tied to the surface of the Earth) is aligned to be **perpendicular** (parallel?) to the meridian. Since this is not an inertial frame, the two weights experience a centrifugal force, apart from the gravitational one.

The gravitational force on the surface of the Earth is

$$\vec{F}_g = -G_N M_{\oplus} M_{\text{grav}} \frac{\vec{r}}{r^3}, \quad (11.1)$$

while the centrifugal force is

$$\vec{F}_{\omega} = M_{\text{inert}} \omega^2 \left( \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega} \right), \quad (11.2)$$

where  $\omega$  is Earth's angular velocity, and  $r$  is the Earth's radius.

- Convince yourself of the validity of these two formulas. Draw the two forces acting on objects in the torsion balance of Figure 1.
- The main idea of the experiment was to check whether  $\frac{M_{\text{inert}}^{(1)}}{M_{\text{grav}}^{(1)}} = \frac{M_{\text{inert}}^{(2)}}{M_{\text{grav}}^{(2)}}$  for different materials. If this were true, then the equivalence principle holds. (The actual value of the ratio is somewhat ambiguous and depends on the definition. The value is fixed to be 1 by choosing the appropriate Newton's constant.) Show that the measurable misalignment between the forces  $\vec{F}^{(i)} = \vec{F}_g^{(i)} + \vec{F}_{\omega}^{(i)}$  acting on two objects is

$$\alpha = \frac{|\vec{F}^{(1)} \times \vec{F}^{(2)}|}{|\vec{F}^{(1)}| |\vec{F}^{(2)}|} = \left| \frac{M_{\text{inert}}^{(1)}}{M_{\text{grav}}^{(1)}} - \frac{M_{\text{inert}}^{(2)}}{M_{\text{grav}}^{(2)}} \right| \frac{(\vec{\omega} \cdot \vec{r}) |\vec{\omega} \times \vec{r}| r}{G_N M_{\oplus}}. \quad (11.3)$$

Assume that the gravitational force is much stronger than the centrifugal one.

- Using the result of the previous point, what would happen if somebody carried out the Eötvös experiment at the equator? What is the most favourable latitude to carry out the experiment (namely, at which latitude the possible misalignment would be largest)?

**SOLUTION: (a)** The form of the gravitational forces (11.1), where  $M_\oplus$  denotes Earth's mass, follows from the following considerations. If one assumes that Earth is a perfect ball of radius  $r = \|\vec{r}\|$  filled with a material of homogeneous density, then Birkhoff's theorem applies (it is better to call it the Jebsen-Birkhoff theorem), and the force can be calculated by assuming that all mass is concentrated at the center. Even if the above assumptions are not completely correct (and they are not since there are deviations from the above assumptions), they will not affect the experiment since the size of the torsion balance is too small to probe inhomogeneities or irregularities of Earth – the effect of these will be in changing  $M_\oplus$  to a slightly different value, which is however identical for both test masses.

The second force (11.2) is the centrifugal force,  $\vec{F}_{\text{cf}} = -M_{\text{inert}}\vec{\omega} \times (\vec{\omega} \times \vec{r})$  induced by the rotation of the Earth (and therefore also by the rotation of the laboratory frame), where  $\vec{\omega} = \omega\hat{z}$  is Earth's rotation vector (pointing from the center towards its north pole). By making use of the identity,  $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$ , one immediately obtains Eq. (11.2). The two forces are shown in figure 2.

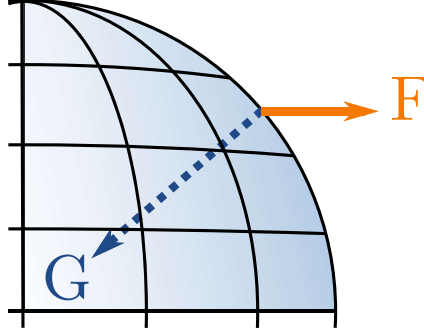


Figure 2: The gravitational force  $G$  points towards the center of the Earth, while the centrifugal force  $F$  (depicted in red) points in the direction  $\hat{r} - \cos(\theta)\hat{\omega}$  ( $\hat{r} = \vec{r}/r$ ,  $\hat{\omega} = \vec{\omega}/\omega$  and  $\theta = \angle(\vec{\omega}, \vec{r})$ ), which is orthogonal to the axis of rotation  $\vec{\omega}$  and lies in the equatorial plane.

**(b)** The first (important) observation is that the size of the rod of the torsion balance can be neglected. After this assumption, it is not hard to see that the two forces can be written as,

$$\begin{aligned}\vec{F}^{(1)} &= -\frac{G_N M_\oplus M_{1,\text{grav}}}{r^3} \left[ \vec{r} - \frac{r^3 \omega^2}{G_N M_\oplus} \frac{M_{1,\text{inert}}}{M_{1,\text{grav}}} \left( \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega} \right) \right] \\ \vec{F}^{(2)} &= -\frac{G_N M_\oplus M_{2,\text{grav}}}{r^3} \left[ \vec{r} - \frac{r^3 \omega^2}{G_N M_\oplus} \frac{M_{2,\text{inert}}}{M_{2,\text{grav}}} \left( \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega} \right) \right].\end{aligned}\quad (11.4)$$

Now the magnitude of the two forces can be approximated by,  $\|\vec{F}^{(i)}\| \simeq G_N M_\oplus M_{i,\text{grav}}/r^2$  and their vector product by,

$$\begin{aligned}\vec{F}^{(1)} \times \vec{F}^{(2)} &= \frac{\|\vec{F}^{(1)}\| \|\vec{F}^{(2)}\|}{r^2} \left[ \vec{r} - \frac{r^3 \omega^2}{G_N M_\oplus} \frac{M_{1,\text{inert}}}{M_{1,\text{grav}}} \left( \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega} \right) \right] \\ &\quad \times \left[ \vec{r} - \frac{r^3 \omega^2}{G_N M_\oplus} \frac{M_{2,\text{inert}}}{M_{2,\text{grav}}} \left( \vec{r} - \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{\omega} \right) \right] \\ &\simeq \|\vec{F}^{(1)}\| \|\vec{F}^{(2)}\| \frac{r \omega^2}{G_N M_\oplus} \left[ \frac{M_{2,\text{inert}}}{M_{2,\text{grav}}} - \frac{M_{1,\text{inert}}}{M_{1,\text{grav}}} \right] \frac{(\vec{\omega} \cdot \vec{r})}{\omega^2} \vec{r} \times \vec{\omega}.\end{aligned}\quad (11.5)$$

Upon inserting this into Eq. (11.3), one gets the desired result.

(c) From Eq. (11.3) we see that the misalignment is maximized when the product  $|\vec{\omega} \cdot \vec{r}| |\vec{\omega} \times \vec{r}|$  maximizes. This product can be also rewritten in terms of the polar angle  $\theta$  as (recall that  $\vec{\omega}$  points in the azimuthal direction),

$$\vec{\omega} \cdot \vec{r} |\vec{\omega} \times \vec{r}| = (\omega r)^2 \sin(\theta) \cos(\theta) = \frac{1}{2}(\omega r)^2 \sin(2\theta), \quad (11.6)$$

which maximizes when  $\theta = \pi/4$  radians (or  $45^\circ$ ). That also means that at the equator the effect would be zero (from figure 2 we see that at the equator the two forces are anti-parallel to each other, such that the effect there vanishes).

## ■ PROBLEM 12 Relativistic particle in an electromagnetic field.

The properties of the electromagnetic field are governed by the vector potential  $A^\mu$ . It is given by

$$A^\mu(x^\nu) = \left( \varphi(x^\nu), \vec{A}(x^\nu) \right).^1 \quad (12.1)$$

The action of a charged particle in the presence of an electromagnetic field is given by

$$S = -mc^2 \int_a^b d\tau - \frac{e}{c} \int A_\mu(x^\nu) dx^\mu, \quad (12.2)$$

where  $d\tau = \sqrt{-\eta_{\mu\nu} dx^\mu dx^\nu}/c$  is the proper time,  $m$  denotes particle's mass and the charge  $e$  of the particle governs its interaction strength with the electromagnetic field given by the vector potential  $A^\mu$ . In this problem, we would like to calculate the equation of motion for this particle, in the presence of the electromagnetic vector potential  $A^\mu$ .

- (a) Write down the variation of the action  $\delta_x S = 0$  with respect to  $x^\nu$  for a neutral particle ( $e = 0$ ). You should first derive

$$\delta d\tau = -\frac{dx_\mu d(\delta x^\mu)}{c^2 d\tau}. \quad (12.3)$$

- (b) Let us now define the following quantity

$$u^\mu = \frac{dx^\mu}{d\tau} = \frac{dx^\mu}{\sqrt{1 - \frac{\vec{v}^2}{c^2}} dt} = \left( \frac{c}{\sqrt{1 - \frac{\vec{v}^2}{c^2}}}, \frac{\vec{v}}{\sqrt{1 - \frac{\vec{v}^2}{c^2}}} \right). \quad (12.4)$$

Evaluate  $u_\mu u^\mu$ .

- (c) Now, Taylor expand  $A_\mu(x^\nu + \delta x^\nu)$  around  $x_\nu$  to find  $\delta A_\mu$ . Furthermore, piecing things together, derive the differential equation that governs the motion of our charged particle whose action is given by (12.2). Give the physical interpretation of the (time-like and space-like) terms obtained by the variation of the second term in Eq. (12.2).

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<sup>1</sup>Note that  $A^\mu \rightarrow A_\mu = \eta_{\mu\nu} A^\nu = (-\varphi, \vec{A})$ .

**SOLUTION: (a)** Varying the action (12.2) gives,

$$\delta_x S = mc^2 \int_a^b \frac{\eta_{\mu\nu} dx^\mu \delta dx^\nu}{c^2 d\tau}. \quad (12.5)$$

Since the two differential operators are independent, we have  $\delta dx^\mu = d\delta x^\mu$ . Next step is to integrate by parts (12.5)

$$\delta_x S = m \int_a^b d \left[ \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \delta x^\nu \right] - m \int_a^b \eta_{\mu\nu} d \left[ \frac{dx^\mu}{d\tau} \right] \delta x^\nu. \quad (12.6)$$

The first term on the right hand side is a boundary term  $[\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \delta x^\nu]_a^b$  which can be set to zero *e.g.* by choosing  $\delta x^\nu = 0$  at the end points  $a$  and  $b$ . Dropping the boundary terms, one gets,

$$\delta_x S = -m \int_a^b \eta_{\mu\nu} \left[ \frac{d^2 x^\mu}{d\tau^2} \right] d\tau \delta x^\nu. \quad (12.7)$$

Now since  $\delta x^\nu$  is arbitrary, we conclude that the following equation of motion must hold for a neutral particle,

$$\frac{d^2 x^\mu}{d\tau^2} = 0. \quad (12.8)$$

Introducing a canonical momentum,  $p^\mu = mu^\mu$ ,  $u^\mu = dx^\mu/d\tau$ , this equation can be also written as,  $dp^\mu/d\tau = 0$ . Parenthetically we note that in non-equilibrium problems it is useful to include the boundary terms to the equation of motion,

$$m \frac{d^2 x^\mu}{d\tau^2} - m \frac{dx^\mu}{d\tau} [\delta(\tau - \tau_b) - \delta(\tau - \tau_a)] = 0. \quad (12.9)$$

**(b)** Firstly, we have  $d\tau = \sqrt{1 - (\frac{d\vec{x}}{cdt})^2} dt$ , from which Eq. (12.4) follows. Next, we have,

$$u_\mu u^\mu = -(u^0)^2 + (\vec{u})^2 = -\frac{c^2}{1 - \frac{\vec{v}^2}{c^2}} + \frac{(\vec{v})^2}{1 - \frac{\vec{v}^2}{c^2}} = -c^2, \quad (12.10)$$

from which we also conclude that,  $p_\mu p^\mu = -m^2 c^2 = -(E/c)^2 + (\vec{p})^2$ , or  $E = \sqrt{(\vec{p}c)^2 + (mc^2)^2} = m\gamma c^2$ .

**(c)** Variation of the second term in action (12.2)

$$\delta_x S_{\text{int}} = -\frac{e}{c} \int_a^b [(\partial_\nu A_\mu(x^\nu)) dx^\mu \delta x^\nu + A_\nu(x^\mu) d\delta x^\nu]. \quad (12.11)$$

Upon integrating by parts the second term one obtains,

$$\delta_x S_{\text{int}} = -\frac{e}{c} \left\{ A_\nu(x^\mu) \delta x^\nu \Big|_a^b + \int_a^b [\partial_\nu A_\mu(x^\nu) - \partial_\mu A_\nu(x^\mu)] dx^\mu \delta x^\nu \right\}. \quad (12.12)$$

Incorporating this contribution into the equation of motion yields,

$$m \frac{d^2 x^\mu}{d\tau^2} - m \frac{dx^\mu}{d\tau} [\delta(\tau - \tau_b) - \delta(\tau - \tau_a)] = -\frac{e}{c} \eta^{\mu\nu} \left[ F_{\nu\alpha} \frac{dx^\alpha}{d\tau} + A_\nu(x^\mu) [\delta(\tau - \tau_b) - \delta(\tau - \tau_a)] \right]. \quad (12.13)$$

where  $F_{\nu\mu} = \partial_\nu A_\mu(x^\nu) - \partial_\mu A_\nu(x^\mu)$  is the electromagnetic field strength and for completeness we also included the boundary terms. Note that the change in canonical momentum  $P^\mu = m \frac{dx^\mu}{d\tau} - \frac{e}{c} \eta^{\mu\nu} A_\nu$  between the two boundaries also contributes to the motion of a (charged) particle. Recall that canonical momentum is defined as a variation of the action with respect to  $dx^\mu/d\tau$ , *i.e.*  $P_\mu = \delta S / \delta(dx^\mu/d\tau)$ . The equation of motion can be broken into the time-like and space-like components, to give,

$$m \frac{d^2 x^0}{d\tau^2} = \frac{e}{c} F_{0i} \frac{dx^i}{d\tau}, \quad m \frac{d^2 x^i}{d\tau^2} = -\frac{e}{c} \left( F_{i0} \frac{dx^0}{d\tau} + F_{ij} \frac{dx^j}{d\tau} \right) = e\gamma (E^i + \epsilon^{ijl} v^j B^l), \quad (12.14)$$

where we ignored the boundary terms and we made use of use,  $v^j = dx^j/dt$ ,  $F_{i0} = -E^i$ ,  $F_{ij} = -\epsilon^{ijl} B^l$ . These equations can be also written more compactly as,

$$m \frac{d^2 x^0}{dt^2} = -\frac{e}{c} \vec{E} \cdot \vec{v}, \quad m \frac{d^2 \vec{x}}{dt^2} = e \left( \vec{E} + \vec{v} \times \vec{B} \right). \quad (12.15)$$

The second equation is the usual Lorentz force, while the first (time-like) equation tells us that the time will exhibit a dilatation in presence of an electric field that is parallel to the motion of a particle.

For your homework, derive the Hamiltonian for a relativistic charged particle. (This problem is a bit tricky, because this is a constrained system, *i.e.*  $P_\mu P^\mu = -m^2 c^2$  must be satisfied. You must take account of that when you perform a Legendre transformation.)